Valvular Heart Disease 1

Valvular heart disease: from mechanisms to management

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Summary

Valvular heart disease is common and its prevalence is rapidly increasing worldwide. Effective medical therapies are insufficient and treatment was historically limited to the surgical techniques of valve repair or replacement, resulting in systematic underprovision of care to older patients and those with substantial comorbidities, frailty, or left ventricular dysfunction. Advances in imaging and surgical techniques over the past 20 years have transformed the management of valvular heart disease. Better understanding of the mechanisms and causes of disease and an increasingly extensive and robust evidence base provide a platform for the delivery of individualised treatment by multidisciplinary heart teams working within networks of diagnostic facilities and specialist heart valve centres. In this Series paper, we aim to provide an overview of the current and future management of valvular heart disease and propose treatment approaches based on an understanding of the underlying pathophysiology and the application of multidisciplinary treatment strategies to individual patients.

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Introduction

Valvular heart disease is caused by the malfunction of one or more heart valves. This common cardiovascular condition affects 5–10% of people aged 65–74 years and 10–20% of people older than 75 years, and its prevalence is rapidly increasing, particularly in high-income countries.[1,2] While changing societal demographics and an ageing population are leading to a growing incidence of primary (or degenerative) diseases, improved therapeutic options and the subsequent prolonged survival of patients with heart failure contribute to a corresponding increase in secondary (or functional) manifestations of disease.

Better understanding of the mechanisms and causes of valvular heart disease and an increasingly extensive and robust evidence base provide a platform for the delivery of individualised treatment by multidisciplinary heart teams working within networks of diagnostic facilities and specialist heart valve centres.[3] Effective medical therapies for valvular heart disease are scarce and treatment was historically limited to the surgical techniques of valve repair or replacement, resulting in a systematic underprovision of care to older patients and to those with substantial comorbidities, frailty, or left ventricular dysfunction.[4,5] This paradox—in which the populations at greatest risk of negative outcomes from disease were the least likely to receive treatment—has been addressed during the past 20 years by advances in imaging and by the rapid emergence of transcatheter techniques that have enabled clinicians, researchers, engineers, device manufacturers, and investors to transform the management of valvular heart disease. Transcatheter techniques are already addressing the needs of thousands of patients who are ineligible for surgery, and the short-term (and potential mid-term) clinical advantages of these techniques—avoiding general anaesthesia, cardiopulmonary bypass, and prolonged hospital stay—have now been substantiated in a series of landmark clinical trials,[6–8] enabling the progressive expansion of their use to younger patients and to those at lower surgical risk.

In this Series paper, we aim to provide an overview of the current and future management of valvular heart disease and propose treatment approaches based on an understanding of the underlying pathophysiology and application of multidisciplinary treatment strategies to individual patients.

Pathophysiology-guided medical treatment

Valvular heart disease can be subdivided into two categories on the basis of aetiology: primary (or degenerative) disease, if caused by dysfunction of a valve itself, and secondary (or functional) disease, if a result of underlying myocardial disease. However, research during the past 5 years has questioned this classification, which does not sufficiently account for actual disease triggers (particularly in patients with secondary valve lesions) or morphological consequences on the valve itself and the cardiac chambers.[9,10]

Although knowing the cause of valvular heart disease is essential to clinical and medical management, understanding the mechanisms of valve failure also has crucial technical implications for the feasibility of valve repair and might determine the eventual need for valve replacement altogether (figure 1).

Primary disease

Primary valvular heart disease includes stenosis or regurgitation caused by degenerative changes (eg, calcification, chordal rupture, or myxomatous infiltration) or, less frequently, by infection (ie, rheumatic heart disease or endocarditis) or congenital malformations (eg, bicuspid aortic valve or Barlow's mitral valve disease). Medical treatments are currently unable to target the underlying causes of the valve lesions and therefore have small utility in primary

valvular heart disease, with the notable exception of infective endocarditis, for which appropriate antibiotic therapy is essential as a stand-alone treatment or in combination with surgery.[11] Antibiotic prophylaxis is important for patients who are at risk of developing rheumatic heart disease or who have signs of secondary cardiac involvement.[12]

Attempts to delay or prevent the progression of aortic stenosis using statins[13–15] or antiosteoporotic drugs[16] have been disappointing. Lipoprotein(a) has been shown to have a causal role in the pathophysiology of aortic valve calcification and progression of aortic stenosis and could be a novel therapeutic target.[17,18] Tafamidis, a transthyretin stabiliser, is considered in addition to valve replacement for selected patients with aortic valve pathology associated with cardiac amyloidosis.[19] Although angiotensin-converting enzyme inhibitors (and other vasodilators) can ameliorate the deleterious maladaptive myocardial response to aortic stenosis and alleviate symptoms of valve regurgitation, no medical approach has yet been shown to affect the natural progression of primary valvular heart disease. Consequently, and importantly, none of these treatments should delay referral for surgical or transcatheter intervention if indicated. Additional research efforts are sorely needed to improve understanding of the genetic and molecular mechanisms underlying calcific valve disease and to enable further identification of potential pharmacological targets,[20] which are addressed in more detail in the second paper in this Series.

Secondary disease

Secondary valvular heart disease typically affects the mitral and tricuspid valves due to their upstream positions in the heart, and results from an imbalance between closing and tethering forces related to one or more of the following factors: alterations of ventricular geometry, displacement of the papillary muscles, and interventricular dysynchrony.[21] Notably, an inferobasal myocardial infarction can cause posterior mitral leaflet insufficiency, which can lead to secondary mitral regurgitation despite normal left ventricular size and function. Whereas these pathophysiological mechanisms imply focal or global myocardial dysfunction (of dilated or ischaemic origin), annular dilatation secondary to chronic atrial fibrillation or elevated left atrial pressure due to diastolic dysfunction is increasingly recognised as a separate entity (known as atrial secondary mitral or tricuspid regurgitation)[10,22] and is associated with better overall prognosis.[23] The

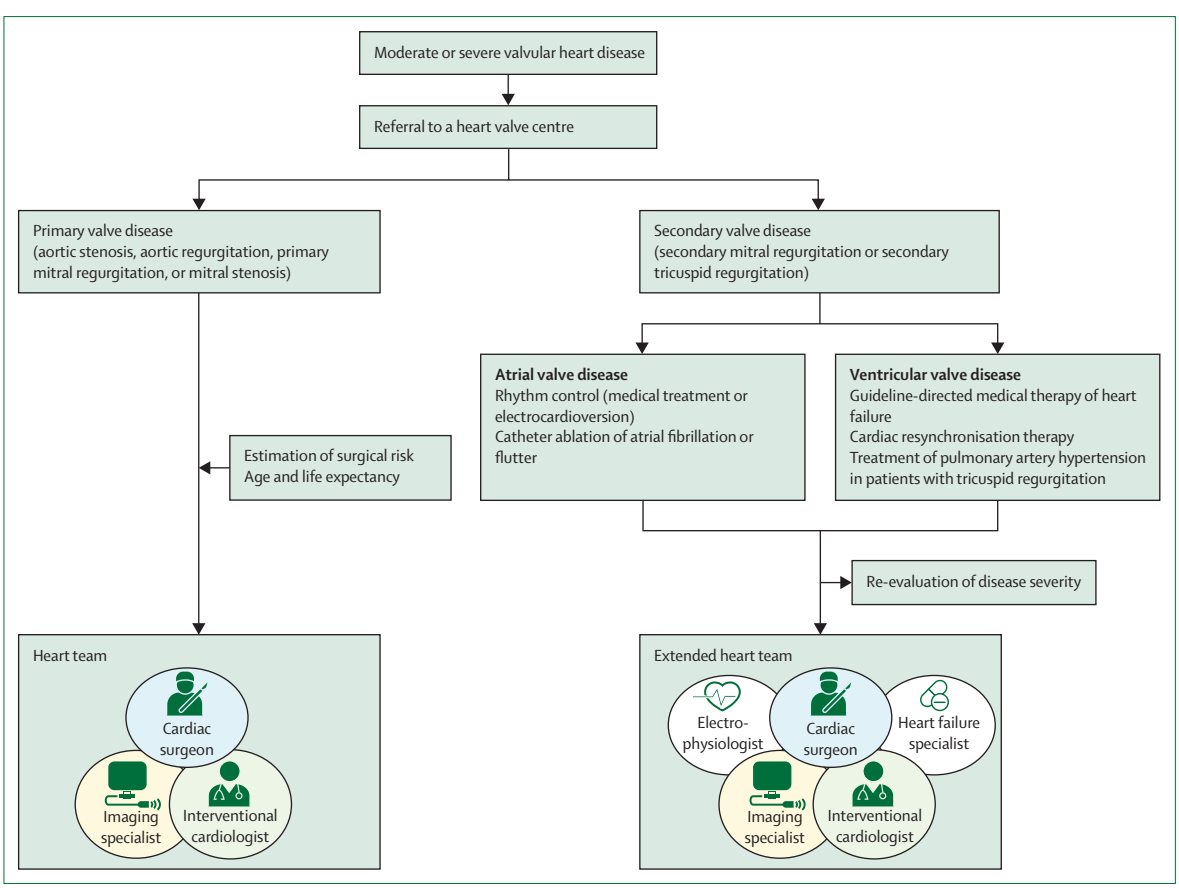


Figure 1: **Diagrammatic representation of clinical management pathways for patients with valvular heart disease**

distinction between these pathophysiological mechanisms is key because the medical management of secondary mitral regurgitation differs fundamentally according to the underlying cause (figure 1). Furthermore, mixed pathophysiological mechanisms could be implicated when cardiac remodelling associated with secondary mitral or tricuspid regurgitation increases the likelihood of atrial fibrillation and annular dilatation.

Medical therapy for secondary mitral regurgitation related to left ventricular dysfunction follows the general principles of treatment for heart failure with reduced ejection fraction.[24] The severity of secondary mitral regurgitation is reduced by β blockers[25] and sarcubitril–valsartan (in association with reverse ventricular remodelling[26]). In the PRIME study,[26] which mainly enrolled patients with non-severe chronic secondary mitral regurgitation, the effective regurgitant orifice area was reduced by 30% with sacubitril–valsartan, compared with 9% with valsartan alone. The combination and up-titration of a β blocker, renin–angiotensin system inhibitor, and mineralocorticoid receptor antagonist reduced secondary mitral regurgitation in 40% of patients in one observational study,[27] although the potential value of adding SGLT2 inhibitors remains to be investigated in this context. Nevertheless, irrespective of the initiated therapy, secondary mitral regurgitation remains unchanged in about 60% of patients despite optimal medical treatment.[26,27] Furthermore, most patients with secondary mitral regurgitation are unable to tolerate maximum medical treatment. Of the patients included in the COAPT trial, which compared transcatheter edge-to-edge repair in combination with medical treatment with medical treatment only, 39% received three classes of medication and only 2% tolerated maximum treatment doses.[28] Cardiac resynchronisation therapy should be considered for patients who fulfil current indications, even if the magnitude of reduction of mitral regurgitation is relatively small.[29–31]

Other than diuretics, no specific medical treatment is effective in reducing secondary tricuspid regurgitation, although vasodilators can be considered for patients with precapillary pulmonary hypertension.

The effect of medical treatment on atrial secondary mitral and tricuspid regurgitation is largely unknown, although the severity of regurgitation can be reduced by rhythm-control strategies, including electrical cardioversion[32,33] or surgical or catheter pulmonary vein isolation.[33,34]

Finally, and importantly, the severity of secondary mitral and tricuspid regurgitation should be re-evaluated after the initiation and up-titration of medical treatment and before considering an intervention (figure 1). Any reduction in regurgitation might only be temporary and frequent clinical and imaging follow-up is needed.

Intervention timing and method: the multidisciplinary heart team

The increasing variety and complexity of innovative diagnostic and therapeutic options for valvular heart disease have created the need for interdisciplinary heart teams working in specialist heart valve centres. Such teams are supported by a network of referring physicians, who have an essential role in the detection of disease through routine clinical examination (in particular auscultation) and careful echocardiographic assessment.[35] Requirements for the specialist heart valve centre include institutional cardiology and cardiac surgery departments providing a 24 h per day, 7 days per week service; involvement of clinical and interventional cardiologists, cardiac surgeons, imaging specialists, and cardiovascular

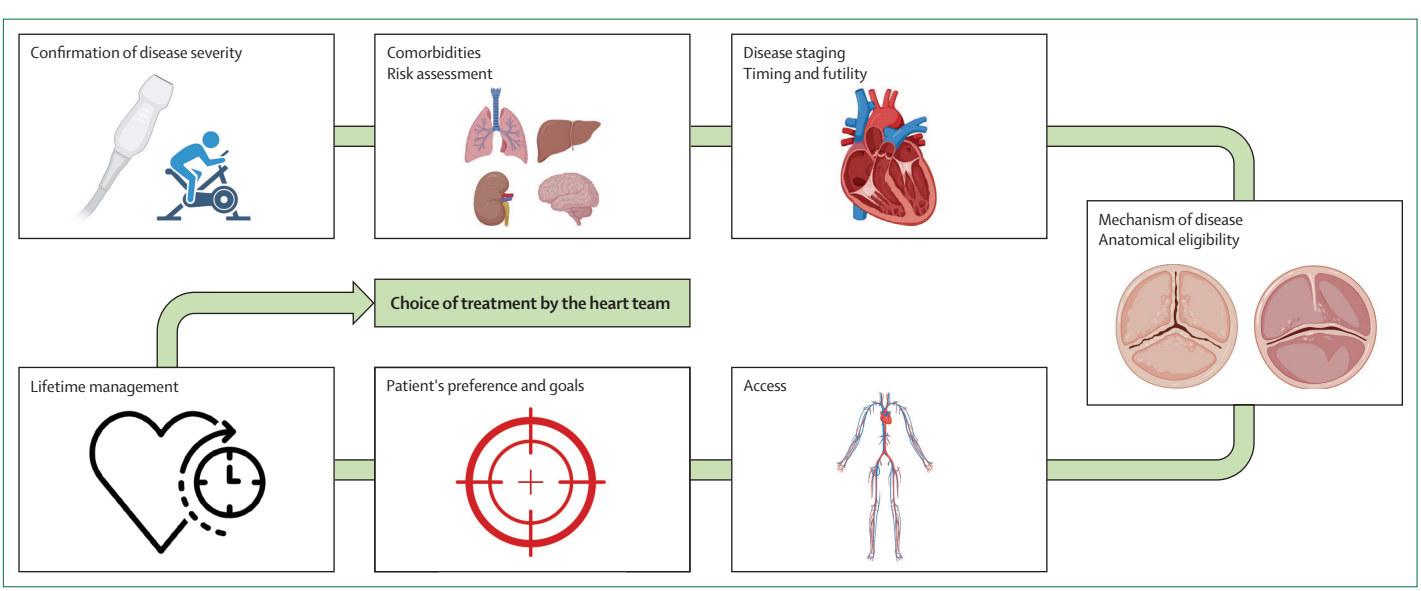


Figure 2: **Key steps for clinical decisions by the multidisciplinary heart team**

anaesthetists supported by additional specialists, as required, according to disease pathophysiology and clinical context (figure 1); and appropriate infrastructure and programmes for education and quality control. The role of the heart team in this shared decision-making model is to interpret information from international guidelines, prospective randomised trials, and observational studies, and apply it to each patient's condition, considering their comorbidities and anticipated life expectancy. The risks and benefits of the available treatment options should be considered alongside the likely durability of the outcome, changes in lifestyle and medication, method of treatment (surgical or transcatheter intervention), and the patient's personal preferences and goals of care (figure 2).

The optimal timing of intervention is of fundamental importance. Many studies have shown considerable improvement in the outcomes of both surgical and transcatheter interventions over the past decade.[36,37] Although such improvement relates partly to technical refinements and increasing operator experience, earlier referral to heart valve centres has been an essential part of this progress. Staging of valvular heart disease has been proposed as a more pertinent approach to define the progressive evolution of disease associated with poor prognosis and increasing rates of cardiovascular events, to determine the timing of treatment, and to provide a better indicator of medical futility. The associated concept of cardiac damage as a consequence of single or combined valvular heart disease has been derived and validated for aortic stenosis,[38,39] mitral regurgitation,[40,41] and tricuspid regurgitation[42,43] and, in this context, advanced imaging methods (particularly cardiac magnetic resonance) that enable the detection of myocardial fibrosis[44] and the precise assessment of right ventricular function[45] are increasingly important.

Panel 1: **Surgical innovations for the treatment of valvular heart disease**

Aortic valve disease
- Rapid-deployment and sutureless valves
- Totally endoscopic surgery
- Robotic surgical platforms
- Hemisternotomy or right anterior minithoracotomy
- Valve repair for aortic regurgitation
- More frequent use of the Ross procedure
- Novel leaflet treatment improving tissue valve durability

Mitral valve disease
- Right lateral minithoracotomy
- Totally endoscopic surgery
- Robotic surgical platforms

Tricuspid valve disease
- Right lateral minithoracotomy
- Early patient referral
- Dedicated risk assessment (eg, by TRI-SCORE)
- Liberal use of concomitant tricuspid valve repair during left-sided cardiac surgery

Interventions in calcific and rheumatic valve disease: replacement as a default strategy

Surgery for the treatment of aortic stenosis: established and still evolving

With few exceptions (eg, exposure to chest radiotherapy or underlying cardiac amyloidosis), leaflet calcification is the principal mechanism underlying the development of aortic stenosis. Mechanical surgical aortic valve replacement is the default strategy for younger patients (<65 years) and those with stenosis affecting a congenital bicuspid valve. Mechanical valves have lifelong durability; however, their use requires anticoagulation therapy with vitamin K antagonists, leading to an increased risk of stroke or bleeding compared with biological valves. Nevertheless, long-term (≥15 years) results are very positive, with freedom from valve-related mortality in an estimated 93% of patients (freedom from thrombo-embolic events 86%, bleeding events 81%, and valve thrombosis 99%).[46] Benefits of mechanical compared with biological valves are particularly pronounced in patients aged 45–54 years but are less striking in those aged 55–64 years.[47] Although an imperfect surrogate for life expectancy, age therefore remains an important determinant for valve selection.

Current surgical management of aortic stenosis is increasingly focused on the reduction of overall physiological stress through the use of less invasive surgical incisions, which reduce blood loss and frequency of new-onset atrial fibrillation, accelerate recovery, and increase the acceptability of surgical intervention to patients[48] (panel 1). A hemisternotomy or right anterior minithoracotomy via the second or third intercostal space[49] is largely used for surgical aortic valve replacement nowadays. In addition, increasing numbers of centres have already adopted total endoscopic aortic valve replacement using new technologies, including three-dimensional (3D) endoscopes and monitors, voice-controlled endoscopic cameras, robotic surgical platforms, automated suturing devices, rapid deployment and sutureless valves, and conventional biological and mechanical valves adapted for minimally invasive surgery.[50]

Severe patient–prosthesis mismatch is associated with increased morbidity and mortality,[51] and root enlargement should always be considered in patients (particularly women) with a small aortic root and those with a BMI greater than 30 kg/m². However, this additional procedure increases total bypass and aortic cross-clamp times[52] and is therefore rarely used in older patients (>75 years), those with impaired left ventricular function, and during reoperations (particularly if a prosthetic mitral valve is already in place).

Despite its relative complexity and the high levels of training required to perform the operation, the Ross

procedure—a pulmonary autograft in the aortic position combined with bioprosthetic pulmonary valve replacement—is a suitable alternative for patients younger than 65 years. The patient's own pulmonary valve is a more durable option than any biological valve and the overall risk of complications is reduced, in particular because there is no need for lifelong anticoagulation therapy. A 2022 study in patients with a median age of 16 years (IQR 13–30) showed an overall survival of 99% and rates of right ventricular outflow tract reoperation of only 6% at the 10-year follow-up.[53] Similarly, longer-term studies confirm a mean 25-year survival rate of 88% (SD 5) after the Ross procedure.[54,55]

Transcatheter aortic valve implantation: a disruptive technology

Surgical aortic valve replacement and transcatheter aortic valve implantation are complementary treatment options for aortic stenosis. Randomised controlled trials comparing the two interventions show that transcatheter aortic valve implantation is superior to medical therapy in patients at extreme risk[56] and results in similar survival to surgery at 5-year follow-up in patients at high,[57–60] intermediate,[61–67] and low[6,68–71] risk. Importantly, patients in these seminal studies were older adults (usually >75 years) and those with adverse anatomical characteristics for either procedure (including a bicuspid aortic valve or coexistent coronary, valve, or aortic root disease) were excluded. Rates of vascular complications, pacemaker implantation, and paravalvular regurgitation are higher after transcatheter aortic valve implantation whereas severe bleeding, acute kidney injury, and new-onset atrial fibrillation are more frequent after surgical aortic valve replacement.[67] Most patients undergoing transcatheter aortic valve implantation have a fast recovery and short hospital stay and can rapidly return to normal activities—key short-term factors that could influence patient choice.

According to the latest European guidelines, published in 2021, transfemoral transcatheter aortic valve implantation is the primary method of treatment for aortic stenosis in all patients aged 75 years or older.[35] This represents a heterogeneous group, and the most appropriate method of intervention for an individual patient should be decided by a heart team, accounting for key clinical, anatomical, and procedural factors (figure 2, table). Considerations should include the relative risks of surgical aortic valve replacement and transcatheter aortic valve implantation and their long-term outcomes, the feasibility of transfemoral transcatheter aortic valve implantation (and alternative approaches in the small proportion [<10%] in whom transfemoral access is not feasible),[72] and local experience and outcome data (table). These factors should be discussed with the patient (and their family) to allow an informed choice of treatment.

The cause of disease and a detailed anatomical assessment of the aortic valve and root are also important for treatment selection (figure 2). Although much remains to be learned regarding optimal device sizing and procedural technique, transcatheter aortic valve implantation is feasible for patients with congenital or acquired functional bicuspid anatomy at high surgical risk. However, the increased risks of relevant paravalvular leak (which are reduced using newer-generation devices), stroke, conversion to surgery, and second valve implantation need to be balanced against the advantages of a less invasive strategy.[73,74]

Transcatheter balloon aortic valvuloplasty can be considered for very selected patients presenting with cardiogenic shock due to decompensated aortic stenosis (although direct transcatheter aortic valve implantation is emerging as a viable alternative in this setting[75]) or for those who require urgent, high-risk, non-cardiac surgery.[76]

	Transcatheter aortic valve implantation preferable	Surgical aortic valve replacement preferable
Transfemoral approach feasible	Yes	No
Transfemoral approach challenging but surgical aortic valve replacement feasible	No	Yes
Alternative access feasible; increased surgical risk	Yes	No
Sequelae of chest radiation	Yes	No
Porcelain aorta	Yes	No
High likelihood of severe patient–prosthesis mismatch (aortic valve area <0·65 cm^2/m^2 body surface area)	Yes	No
Severe chest deformation or scoliosis	Yes	No
Annular dimensions unsuitable for current transcatheter aortic valve implantation devices	No	Yes
Bicuspid aortic valve	No	Yes
Severe left ventricular outflow tract or leaflet calcification	No	Yes
High risk of coronary artery obstruction	No	Yes
Thrombus in aorta or left ventricle	No	Yes

Adapted from the European Society of Cardiology and European Association for Cardio-Thoracic Surgery guidelines for the management of valvular heart disease.[35]

Table: **Anatomical and procedural factors that guide choice of treatment methods for isolated aortic stenosis**

Lifetime management of severe aortic stenosis

The increased use of bioprosthetic valves in patients older than 50 years has, alongside the rapid emergence of transcatheter options, stimulated renewed focus on the lifetime journey of patients with aortic stenosis, who might require repeated valve interventions with optimal sequencing and appropriate planning (panel 2). These considerations are particularly important for younger patients, who could face the prospect of three or four interventions—especially those with adult congenital heart disease (who might have already had surgery or transcatheter intervention in infancy or childhood) and women of childbearing age (who might wish to avoid the risks associated with mechanical valves and the subsequently necessary anticoagulation therapy during pregnancy).

Panel 2: **Key considerations for lifetime management of severe aortic stenosis**

- Age and life expectancy (which influence the likelihood of future interventions)
- Previous aortic valve interventions (ie, surgical aortic valve replacement, transcatheter aortic valve implantation, or Ross procedure)
- Patient and physician acceptance of the need for anticoagulation therapy
- Concomitant cardiac disease
- Aortopathy
- Other valve lesions
- Coronary artery disease
- Atrial fibrillation
- Comorbidities and frailty (which affect the feasibility of surgery)
- Aortic valve and root anatomy
- Bicuspid or tricuspid aortic valve
- Low coronary ostia (which affects the likelihood of occlusion related to transcatheter aortic valve implantation)
- Annular dimensions (which affects the likelihood of patient–prosthesis mismatch)
- Feasibility of future valve-in-valve procedures
- Likelihood of the need for permanent pacemaker implantation
- Future coronary access

Valve-in-valve transcatheter aortic valve implantation in degenerated surgical aortic bioprostheses is a routine procedure in many centres and short-term outcomes compare favourably with repeat surgical aortic valve replacement.[77] By contrast, successive transcatheter aortic valve implantation (so-called TAV-in-TAV or revalving) procedures are still rarely needed (in an estimated 0·3% of patients) and generally take place within 3–6 years[78] after the index procedure. Periprocedural complications after successive transcatheter aortic valve procedures are infrequent (stroke 1·4%; valve malposition 3·3%; coronary obstruction 0·9%; need for new permanent pacemaker 9·6%); however, although appealing, the notion that successive transcatheter aortic valve implantations can be conducted approximately every 10 years throughout a patient's life is overly simplistic for several reasons. First, the long-term durability of current implantation devices remains unknown and, although data from randomised trials and international registries provide reassuring data for up to 10 years,[79] 15-year outcomes are required for comparison with those of some contemporary surgical bioprosthetic valves.[80] Second, patients with bicuspid valves were excluded from the pivotal trials of randomised transcatheter aortic valve implantation, and this anatomical configuration is more frequent in younger cohorts with aortic stenosis. Third, successive device implants can cause progressive patient–prosthesis mismatch, present technical challenges to re-access the coronary arteries, and might be associated with an increased risk of valve or coronary thrombosis. Finally, surgical excision of transcatheter aortic valves (if needed) is frequently difficult, with a high incidence of adverse events, in particular aortic wall injury.[81] Careful assessment using imaging techniques (eg, CT[82]) and discussion among the multidisciplinary heart team concerning specific clinical and anatomical factors are therefore essential to reach the optimal lifetime outcome for individual patients with severe aortic stenosis.[83]

Treatment of calcific and rheumatic mitral valve disease: synergistic techniques for a diverse disease

Calcific mitral valve disease and mitral annular calcification are age-dependent. Mitral annular calcification has a prevalence of 9–15% in the general population but is associated with relevant mitral valve dysfunction in only a minority of patients (ie, an estimated 25% of patients with mitral annular calcification have mitral stenosis and less than 20% have mitral regurgitation).[84] These patients are at high risk of negative outcomes from any intervention, including surgical repair or replacement, transfemoral transcatheter mitral valve implantation, or hybrid procedures.[85] A prospective study investigating implantation of a balloon-expandable transcatheter aortic valve in patients with mitral annular calcification reported a 1-year mortality of 34·5%.[86,87] Implantation of a dedicated transcatheter mitral valve (eg, the Tendyne valve) could therefore be a safer and more effective treatment option, but further research is needed to support this hypothesis[88,89] (figure 3).

Rheumatic valve disease is rare in high-income countries but remains a major issue in low-income and middle-income countries, affecting more than 30 million people worldwide.[90] Although desirable, surgical valve repair is feasible for only a minority of patients due to the frequent combination of mitral regurgitation and stenosis.[91,92] For patients in whom mitral stenosis is predominant, transcatheter balloon mitral valvuloplasty is the first-line treatment, provided that anatomical characteristics are favourable[93] (figure 3).

Interventions for non-calcific and secondary valve disease: more repair, less replacement

Treatment of severe aortic regurgitation: as early as possible

Multiple mechanisms direct the development of aortic regurgitation, including cusp prolapse, maladaptation due to calcific nodules and leaflet retraction (secondary to rheumatic or degenerative disease), and pure secondary disease due to aortic root dilation (with normal leaflets). Treatment is indicated (even in the absence of symptoms) if there is impaired function (ejection fraction ≤50%) or dilatation of the left ventricle (end-diastolic diameter >70 mm or end-systolic diameter >50 mm [or >25 mm/m²

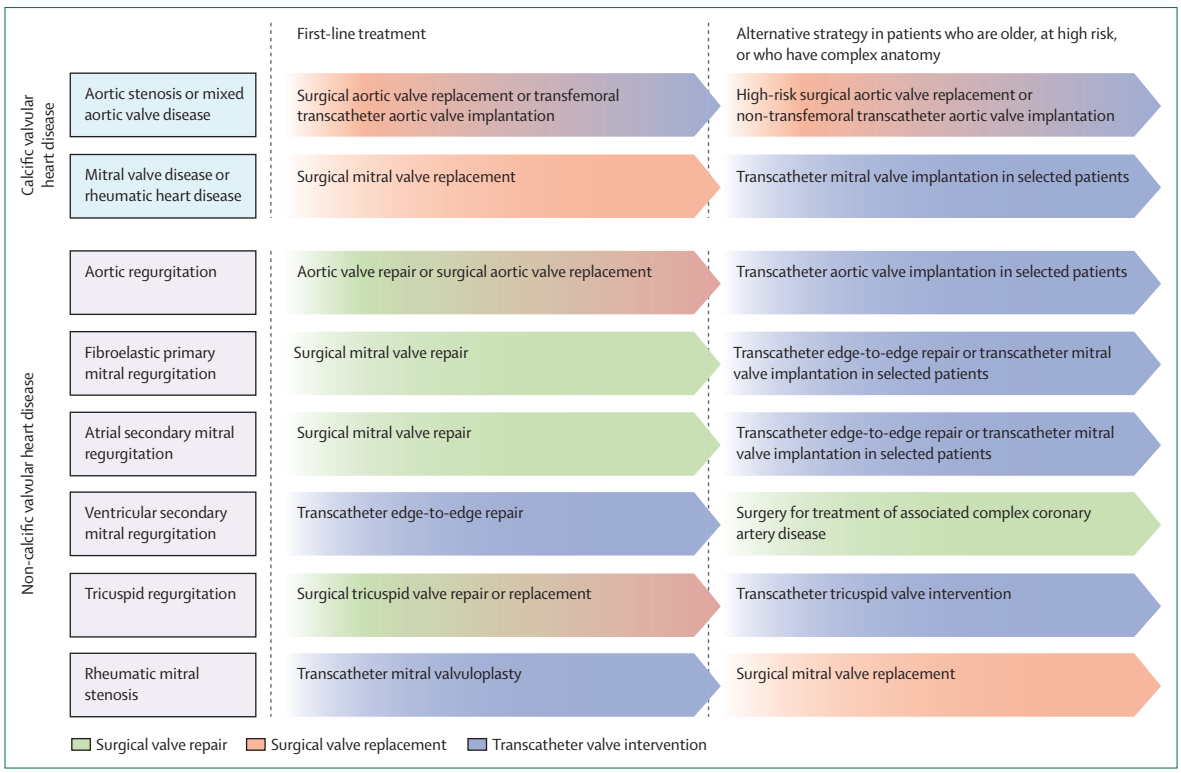


Figure 3: **Management of valvular heart disease according to underlying disease mechanism**

body surface area in patients with a small body size, defined as body surface area <1·68 m²]).[35] About 20% of patients with severe aortic regurgitation have associated aortic stenosis (also called mixed aortic valve disease), and valve replacement should be considered early in this setting.[94] Surgical valve repair is an increasingly used alternative to replacement for the treatment of cusp prolapse or leaflet retraction in experienced centres (figure 3). Mid-term and long-term results (including mean freedom from reoperation of 88% [SD 2] after 5 years and 73% [4] after 10 years) depend on technical aspects of the repair (eg, effective height) and the mechanism of regurgitation.[95] For patients with secondary disease and normal or near-normal cusps, valve-sparing aortic root surgery (known as the David procedure) can be sufficient to correct aortic regurgitation and should be attempted, particularly in patients younger than 65 years.[96]

Although surgical aortic valve replacement remains the first-line treatment for pure aortic regurgitation, off-label transcatheter aortic valve implantation—using both self-expanding and balloon-expandable valves—has been conducted in selected patients at high surgical risk with encouraging outcomes (figure 3). However, a second valve implantation to address prosthesis malposition or relevant residual aortic regurgitation is needed in approximately 10% of patients, even with newer-generation devices.[97] The likelihood of these complications is reduced with a new dedicated transcatheter heart valve system that was approved in the EU in 2021, although the rate of new permanent pacemaker implantation was relatively high (23%).[98] Additional research will be necessary to define the range of application of this and other emerging transcatheter systems dedicated to the treatment of aortic regurgitation.

Surgery for fibroelastic deficiency and secondary mitral regurgitation: unequal effectiveness and supporting evidence

Surgical repair of fibroelastic primary mitral regurgitation and Barlow disease is associated with very favourable immediate and long-term outcomes when conducted in experienced centres[35,99,100] (figure 3). Long-term follow-up data (up to 15 years) show recurrent mitral regurgitation (grade 2+ or higher) in only 13% of patients and a need for reoperation in 5–7%, with excellent survival (66% at 15 years).[101] A minimally invasive approach via right minithoracotomy is now most frequently used in Europe and the USA,[102–104] and further advances—including robotic surgery with 3D high-magnification visualisation using a dual camera scope, multidirectional endoaortic instrument articulation, motion scaling and tremor filtration, and even totally endoscopic procedures[103]—have been introduced in some centres with excellent outcomes[105] (panel 1).

Evidence supporting surgery in ventricular secondary mitral regurgitation is scarce, and such procedures are

rarely done and usually driven by the need for complex surgical revascularisation in patients with ischaemic heart disease[106] (figure 3). By comparison, surgical annuloplasty has shown favourable results for the treatment of atrial secondary mitral regurgitation, with better survival and lower recurrence (residual mitral regurgitation classified as moderate or greater in 7% of patients with atrial *vs* 25% of those with ventricular secondary mitral regurgitation at 2 years).[107]

Transcatheter treatment of the mitral valve: building bridges between established and emerging techniques

Encouraged by the success of transcatheter aortic valve implantation, transcatheter treatment of the mitral valve has attracted increasing attention, technical innovation, and financial investment over the past 10 years. Future development in this field will need to establish the balance between the two fundamental techniques of valve repair and replacement and their respective range of clinical application (panel 3).

Transcatheter mitral edge-to-edge repair is a minimally invasive procedure that is now routinely conducted in several countries. Effectiveness and safety have improved markedly since its introduction as a result of successive device refinements and increasing operator experience,[110,111] although its results remain highly dependent on anatomical characteristics and disease complexity, which are best assessed by 3D echocardiography.[111,112] Target populations, such as older patients (>75 years) with primary[8] or atrial secondary[113] mitral regurgitation who are at high surgical risk and patients with ventricular secondary mitral regurgitation who remain symptomatic despite optimal guideline-directed medical treatment, have been largely undertreated in the past. Whereas the first randomised controlled trial involving transcatheter mitral edge-to-edge repair was designed to compare the procedure to surgery in a mixed population of patients with either primary or secondary mitral regurgitation,[114] subsequent studies have focused on patients with heart failure with ventricular secondary mitral regurgitation,[115] enabling the identification of specific clinical and echocardiographic criteria—derived from the COAPT study[7]—that predict the prognostic response to transcatheter mitral edge-to-edge repair. In selected patients with secondary mitral regurgitation, transcatheter mitral edge-to-edge repair is associated with a 38% reduction in the relative risk of all-cause mortality and a 47% reduction in admission to hospital due to heart failure at a 2-year follow-up (confirmed at 5 years in the crossover population).[116] In addition, successful mitral transcatheter mitral edge-to-edge repair facilitates up-titration of medical treatment in 38% of patients, with associated further reduction of all-cause mortality and hospitalisation due to heart failure.[117] Ongoing randomised trials (REPAIR-MR, NCT04198870; and PRIMARY, NCT05051033) are comparing the outcomes of transcatheter mitral edge-to-edge repair with those of mitral valve surgery in patients with primary mitral regurgitation at intermediate risk.[118]

Transcatheter mitral valve replacement is an additional treatment option that could potentially address the needs of patients with complex anatomy (mainly of primary cause) for whom a suboptimal outcome from transcatheter mitral edge-to-edge repair is anticipated (figure 3). Only one system (using transapical access) is currently certified for use in Europe, but considerable ongoing research efforts are focused on the development of transseptal devices delivered via the femoral vein. Although the exact role of this emerging technology will require further clarification, encouraging results have been obtained, particularly in patients with complex primary mitral regurgitation, mitral annular calcification, or failed rings or clips.[88,89] The principal limitations preventing broader adoption of this technique so far are the risk of left ventricular outflow tract obstruction, thrombotic or bleeding complications, and annular dimensions beyond the size of currently available bioprostheses (panel 3). As seen after surgery, afterload mismatch and delayed patient recovery are common after

Panel 3: **Advantages and disadvantages of transcatheter replacement of the mitral and tricuspid valves, with anatomical criteria favouring replacement**

Advantages of transcatheter valve replacement
- More effective than valve repair in reducing regurgitation
- Durability likely (although largely unknown)

Disadvantages of transcatheter valve replacement
- Higher risk of procedural complications than with valve repair
- Risk of left ventricular outflow tract obstruction (mitral valve)
- Risk of need for new pacemaker implantation (tricuspid valve)
- Interaction with the subvalvular apparatus or right ventricular lead
- Risk of valve thrombosis
- Risk of haemolysis associated with paravalvular leakage

Criteria favouring mitral valve replacement[108]
- Concentric mitral annular calcification with stenosis
- Mitral valve area <3 cm^2
- Clinically relevant mitral stenosis (mean gradient >5 mm Hg)
- Posterior leaflet <5 mm
- Calcification in the grasping zone
- Deep regurgitant cleft
- Leaflet perforation
- Multiple or wide jets
- Rheumatic mitral stenosis
- Failed surgical or transcatheter repair

Criteria favouring orthotopic tricuspid valve replacement[109]
- Large coaptation gap and predominantly central gap
- Severe leaflet tethering
- Tricuspid regurgitation related to cardiac implantable electronic device leads
- Complex leaflet morphology (more than three valve leaflets)
- Leaflet thickening or perforation (eg, rheumatic, carcinoid, or post-endocarditis)
- Previous surgical repair or bioprosthetic valve replacement

valve replacement, although long-term symptomatic improvement and positive remodelling might be more frequent, presumably as a consequence of more effective reduction in mitral regurgitation.[119,120]

Treatment of tricuspid valve disease: at the frontiers of the unknown

Patients with severe tricuspid regurgitation are generally referred too late for treatment and isolated surgery is rarely conducted. Improved disease awareness and earlier intervention are therefore key priorities for emerging surgical and transcatheter programmes to avoid right ventricular failure and subsequent hepatorenal dysfunction, which contribute to poor outcomes.[121] The TRI-SCORE—which was proposed in 2022 and integrates age, burden of heart failure symptoms, renal and hepatic congestion, and biventricular function—performs better than traditional risk scores (which have not been validated for tricuspid surgery) in estimating the risk of in-hospital death[122] and can be applied to patients undergoing first time or repeat isolated tricuspid valve surgery.[123] Consistently with international guidelines, a 2022 randomised trial underscored the importance of concomitant tricuspid valve repair annuloplasty at the time of left-sided valve surgery even in patients with non-severe tricuspid regurgitation and annular dilatation (≥40 mm or >21 mm/m² body surface area), to prevent the later development of severe tricuspid regurgitation.[124] Although tricuspid valve repair was associated with a higher rate of permanent pacemaker implantation in this trial (14% *vs* 3% in the patients without annuloplasty), this finding was not observed in other studies.[125]

Emerging dedicated transcatheter solutions offer valuable options for patients who are ineligible for surgery (figure 3). The anatomy of the tricuspid valve is complex, and multiple technical approaches are likely to be required in the treatment of this condition—including transcatheter edge-to-edge repair, direct annuloplasty, and orthotopic or heterotopic valve replacement.[109] Several observational studies[126–130] have already established the safety and efficacy of these different systems and their ability to improve functional status. Specific considerations associated with transcatheter tricuspid valve replacement are summarised in panel 3. Patient selection is based on clinical and anatomical factors, including rigorous evaluation of right ventricular function (by means of the right ventricular–pulmonary artery-coupling index[131] and right ventricular longitudinal strain[132]), pulmonary pressures and resistance, estimation of surgical risk using dedicated scores,[122,133] and assessment of hepatic and renal function.

The first randomised trial comparing transcatheter correction of isolated tricuspid regurgitation with medical treatment was published in 2023.[134] Using transcatheter edge-to-edge repair, tricuspid regurgitation was reduced to moderate or less in 87% of patients, with very low complication rates (1·7% at 30 days), and was associated with a significant improvement in patient-reported quality of life compared with medical therapy (difference in the Kansas City Cardiomyopathy Questionnaire score of 11·7 points [95% CI 6·8–16·6] between the intervention and control groups), the scale of this effect correlating strongly with the extent of reduction of tricuspid regurgitation. The lack of effect on more objective endpoints (ie, all-cause mortality and hospitalisation attributed to heart failure) could be related to the enrolment of a relatively low-risk population and to the short duration of follow-up (up to 1 year). The results of several randomised trials, expected in 2024, will help to address some of the remaining gaps in evidence.

Multivalvular disease

Multivalvular disease is an underdiagnosed, insufficiently investigated, and undertreated complex condition that is inadequately addressed by international guidelines. Approximately 30–50% of patients with left-sided native valvular heart disease have associated moderate or severe valvular lesions, and their prognosis is poorer than that of patients with single isolated valvular heart disease.[1,135] Moderate or severe mitral regurgitation is observed in 20–30% of candidates for transcatheter aortic valve implantation[136] and more rarely in patients with severe aortic regurgitation (14%).[137] Tricuspid regurgitation is often associated with severe mitral regurgitation (moderate or greater tricuspid regurgitation is seen in 56% of patients undergoing mitral transcatheter edge-to-edge repair)[108] and less frequently with severe aortic stenosis (11–27% of candidates for transcatheter aortic valve implantation).[136] Severe mitral stenosis is found in a small proportion (<3%) of older patients (>75 years) with severe aortic stenosis.[138]

The following diagnostic difficulties can contribute to an underappreciation of disease severity:[139] an increased afterload and elevated intraventricular systolic pressure in moderate or severe aortic stenosis can result in overestimation of the severity of mitral regurgitation; the quantification of aortic regurgitation and mitral stenosis is less reliable in the presence of mitral regurgitation and aortic stenosis; an eccentric mitral regurgitation jet could be mistaken for aortic stenosis; and severe mitral regurgitation or tricuspid regurgitation could be associated with a low-flow or low-gradient state, leading to underestimation of the severity of aortic stenosis and mitral stenosis.

An understanding of the cause and pathophysiology of disease is essential to determine the appropriate sequence of treatment. Surgery is the first-line treatment for all operable patients, because it addresses all valve lesions during a single procedure. By contrast, transcatheter interventions enable a staged approach that allows the effect of correction of the principal valve lesion on remaining valvular heart disease to be assessed, and thereby allows the timing of subsequent interventions to be optimised. Previous studies have reported improvement of secondary mitral regurgitation by at least one grade in more than 50% of patients after

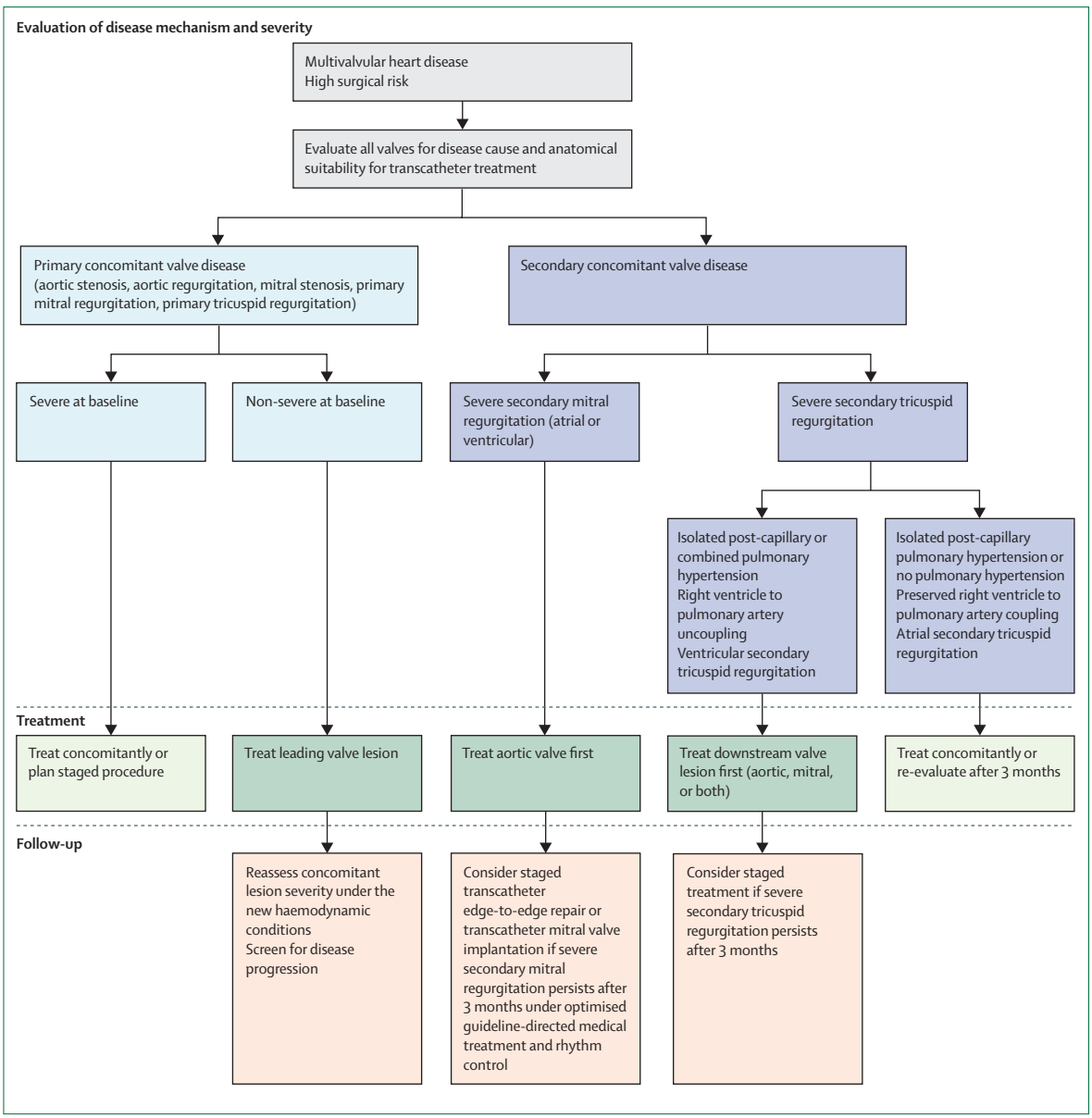


Figure 4: **Proposed strategies for the transcatheter treatment of multiple valvular heart disease according to cause**

transcatheter aortic valve implantation, although improvement strongly depends on disease mechanism and is more frequently observed in secondary than in primary valve disease.[140] Similarly, tricuspid regurgitation improves in 30–40% of patients after treatment of left-sided valvular heart disease.[136] Transcatheter treatment of multiple valvular heart disease should therefore generally target the downstream valve lesion (aortic then mitral, and finally tricuspid) followed by a wait-and-see approach to concomitant secondary disease (figure 4). Data supporting this strategy remain limited to the sequential treatment of mitral and tricuspid regurgitation, in which improved 6-min walking distance and reduced concentrations of N-terminal pro B-type natriuretic peptide translate into reduced rates of readmission to hospital.[141,142] By contrast, concomitant primary valve disease can be treated during the same procedure in selected cases, to reduce invasiveness and the duration of hospital stay.

Conclusions

The management of valvular heart disease is strongly dependent on a sound understanding of the cause and mechanisms of disease, which should be thoroughly assessed in individual patients before consideration of the most appropriate treatment options by a multidisciplinary heart team. This approach is strongly endorsed by current

international guidelines, but requires further validation and support by means of dedicated research. Despite a major surge in the awareness of valvular heart disease and the availability of resources to improve understanding and outcomes of this often-complex condition over the past 20 years, important evidence gaps remain regarding new avenues for medical therapy, the optimal timing of surgical and transcatheter interventions, standardised antithrombotic treatment after mitral and tricuspid procedures, and management of multivalvular disease (including right-sided lesions). Meanwhile, an immediate challenge is to ensure that advances in diagnosis, surgical and transcatheter intervention, and access to treatments that have been established in high-income settings are extended to low-income and middle-income countries, where awareness of valvular heart disease is lower and treatments are largely underused.

Contributors

FP, FB, KH, and BP conceptualised the paper, wrote the original draft, and reviewed and edited the final manuscript.

Declaration of interests

BP reports educational grants from Edwards Lifesciences, paid to his institution (St Thomas' Hospital); advisory board fees from MicroPort and Anteris Technologies; speaker fees from Edwards Lifesciences, Medtronic, and Abbott Laboratories; and has participated on data safety monitoring boards for Valvosoft and Colibri Scientific. FP reports support for attending meetings or travel from Edwards Lifesciences, Medira, Abbott Vascular, Siemens Healthineers, and inQB8 Medical Technologies. All other authors declare no competing interests.

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